

CHAPTER 8

CHOICES

8.1. The Axiom of Choice, AC. For any two sets A, B and any binary relation $P \subseteq (A \times B)$,

$$(\forall x \in A)(\exists y \in B)P(x, y) \implies (\exists f : A \rightarrow B)(\forall x \in A)P(x, f(x)). \quad (8-1)$$

This is the last and most controversial axiom of Zermelo. To understand how such an axiom might be needed, consider the classical example of Russell, where A is a set of pairs of shoes, $B = \bigcup A$ and

$$P(x, y) \iff y \in x.$$

The function

$$f(x) =_{\text{df}} \text{the left shoe in } x, \quad (x \in A)$$

obviously selects a shoe from each pair, in symbols $(\forall x \in A)P(x, f(x))$. If, however, A is a set of pairs of *socks*, then there is no obvious way to *define* a function $f : A \rightarrow \bigcup A$ which selects one sock $f(x) \in x$ from each pair, because (as we stipulate for the example), a pair of socks comprises precisely two perfectly identical objects. We can still prove that a selector function f exists when A is finite, by induction on the number of elements in A (Problem **x8.1**). But in mathematics we can imagine infinite sets of pairs of socks, and in that case we need something like the Axiom of Choice to guarantee the existence of such a function.

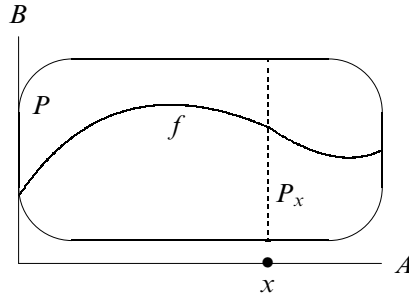
Less amusing but more significant for mathematics is the proof of the basic theorem **2.10**, where we considered a sequence $A_0, A_1, \dots$ of countable sets and began with the phrase

It is enough to prove the theorem in the special case where none of the A_n is empty, in which case we can find for each A_n an enumeration $\pi_n : \mathbb{N} \twoheadrightarrow A_n$.

Perhaps for each n “we can find” (i.e., “there exists”) some enumeration π of A_n , but the rest of the proof needs a *function* ($n \mapsto \pi_n$) which associates a specific enumeration π_n with each n : which of the axioms **(I)** – **(VI)** can be used to prove the existence of such a function? Here $A = \mathbb{N}$, $B = (\mathbb{N} \rightarrow \bigcup_{n=0}^{\infty} A_n)$ and

$$P(n, \pi) \iff \pi : \mathbb{N} \twoheadrightarrow A_n,$$

so that $(\forall n \in \mathbb{N})(\exists \pi \in B)P(n, \pi)$ from the hypothesis that each A_n is non-empty and countable: and the Axiom of Choice guarantees precisely that there

FIGURE 8.1. A selector for $P \subseteq A \times B$.

exists a function $f : \mathbb{N} \rightarrow B$ such that for each $n \in \mathbb{N}$, the value $f(n) = \pi_n$ satisfies $P(n, \pi_n)$, i.e., it enumerates A_n . Such “silent” appeals to the Axiom of Choice, masked by the notation, are very common in mathematics and especially in analysis, where the classical theory of limits and continuous functions cannot be developed in a satisfactory way without choices.

If we picture $P \subseteq A \times B$ as a subset of the product space, then the hypothesis $(\forall x \in A)(\exists y \in B)P(x, y)$ means that the *fiber* or *section*

$$P_x =_{\text{df}} \{y \in B \mid P(x, y)\} \quad (8-2)$$

above each $x \in A$ is non-empty; the Axiom of Choice guarantees the existence of a *selector* for P , a function $f : A \rightarrow B$ which assigns to each $x \in A$ exactly one point in the fiber above it. There are two other, simple reformulations of the axiom which express in different ways the process of “collecting into a whole” any number of unrestricted, non-conflicting choices.

8.2. Definition. A set S is a **choice set** for a family of sets $\mathcal{E}$ if (1) $S \subseteq \bigcup \mathcal{E}$, and (2) for every $X \in \mathcal{E}$, the intersection $S \cap X$ is a singleton. A choice set S *selects* from each $X \in \mathcal{E}$ the unique member of the intersection $S \cap X$.

8.3. Exercise. If $\emptyset \in \mathcal{E}$, then $\mathcal{E}$ does not admit a choice set. Also, if $a \neq b$, then the family $\mathcal{E} = \{\{a\}, \{a, b\}, \{b\}\}$ does not admit a choice set.

8.4. Theorem. The Axiom of Choice is equivalent to the following proposition: every family $\mathcal{E}$ of non-empty and pairwise disjoint sets admits a choice set.

This is the version of **AC** postulated by Zermelo.

PROOF. Assume first the Axiom of Choice and let $U = \bigcup \mathcal{E}$ be the union of the given family of pairwise disjoint, non-empty sets, which means that

$$(\forall X \in \mathcal{E})(\exists x \in U)[x \in X].$$

The Axiom of Choice guarantees that there exists a function $f : \mathcal{E} \rightarrow U$, such that

$$(\forall X \in \mathcal{E})[f(X) \in X];$$

we set $S = f[\mathcal{E}] = \{f(X) \mid X \in \mathcal{E}\}$, and the fact that the members of $\mathcal{E}$ are pairwise disjoint implies easily that S intersects every member of $\mathcal{E}$ in a singleton.

For the converse, assume

$$(\forall x \in A)(\exists y \in B)P(x, y),$$

set for each $x \in A$

$$U_x = \{(t, y) \in P \mid t = x\},$$

and let

$$\mathcal{E} = \{U_x \mid x \in A\}.$$

Each member of $\mathcal{E}$ is non-empty by the hypothesis and it is determined by the constant, first member of the pairs in it, so any two members of $\mathcal{E}$ are disjoint. If S is a choice set for this $\mathcal{E}$, then the function

$$f(x) = \text{the unique } y \text{ such that } (x, y) \in S$$

easily satisfies the conclusion of the Axiom of Choice. $\dashv$

8.5. Definition. A **choice function** for a set A is any partial function

$$\varepsilon : \mathcal{P}(A) \rightarrow A,$$

such that

$$\emptyset \neq X \subseteq A \implies \varepsilon(X) \downarrow \ \& \ \varepsilon(X) \in X.$$

8.6. Lemma. *The Axiom of Choice is equivalent to the assertion that every set admits a choice function.*

PROOF. For every A , obviously $(\forall X \in \mathcal{P}(A) \setminus \{\emptyset\})(\exists y \in A)[y \in X]$; and so, directly from the Axiom of Choice, there must exist some function $\varepsilon : \mathcal{P}(A) \setminus \{\emptyset\} \rightarrow A$ such that

$$(\forall X \in \mathcal{P}(A) \setminus \{\emptyset\})[\varepsilon(X) \in X].$$

The converse is easy enough to leave for an exercise. $\dashv$

8.7. Exercise. *If every set admits a choice function, then the Axiom of Choice is true.*

8.8. But is it true? (1) Naively understood, the Axiom of Choice asserts that if each of a set of non-conflicting choices is possible, then they can all be made independently and their results collected into a completed whole, a set. By this understanding it is quite obvious, it can be justified by the natural interpretation we would give to the Powerset Axiom: when we grant sethood to the class $\{X \mid X \subseteq A\}$ of all subsets of A , we truly mean *all* subsets of A , including those for which the membership criterion is not determined by some explicit law but by free choice, by chance if you will.

The Axiom of Choice is different in form from the earlier “constructive” axioms (II) – (VI), because it *postulates directly* the existence of a set for which it does not supply a definition. Each of (II) – (VI) grants sethood to a specific, explicitly defined collection of objects, it legitimizes a special case of the most appealing (if false) General Comprehension Principle 3.3. *The Axiom of Choice is the only Zermelo axiom other than Extensionality which is not a special case of the General Comprehension Principle.* This is misstated on occasion, to make the claim that the Axiom of Choice is the only one which

demand the existence of objects for which it does not supply a definition, which is not true: the Extensionality and Powerset Axioms do the same, in a more fundamental if indirect manner.

Zermelo introduced the Axiom of Choice explicitly in 1904, in a brief paper in which he used it to prove that every set is well orderable. This was a long-standing conjecture, and Cantor had outlined a proof of it in a letter to Dedekind, then still unpublished. His proof, however (and the related proof of the Cardinal Comparability Hypothesis), depended on intuitions about sets which were not sufficiently explained. In contrast to this, Zermelo made it clear, from the start, that his own detailed proof depended on the Axiom of Choice, and he was immediately attacked for this by some of the leading mathematicians of the time, for introducing a questionable method to derive an implausible conclusion. Given the fact that choice principles were by no means new to mathematics and that they permeate Cantor's earlier reasoning, it is fair to say that the shock was caused more by the realization of the power of the axiom than by its meaning.

In the next, fundamental result, we establish the somewhat surprising equivalence of the Axiom of Choice with two very different set theoretic claims. We have kept the traditional names for these propositions—Axiom, Hypothesis, Theorem—which have been attached to them by the historical accident of when and how they were introduced in the mathematical literature.

8.9. Theorem. *The following propositions are all equivalent.*

- (1) **Axiom of Choice:** *Every set admits a choice function.*
- (2) **Hypothesis of Cardinal Comparability:** *For any two sets A, B , either $A \leq_c B$ or $B \leq_c A$.*
- (3) **Wellordering Theorem:** *Every set is well orderable.*

PROOF. We verify, round-robin style, that each of the first two propositions implies the next, and finally (3) $\implies$ (1).

(1) $\implies$ (2). By Exercise 6.17, the poset $(A \rightharpoonup B)$ of partial injections from A to B is inductive. The idea is to define (using AC) an expansive mapping on $(A \rightharpoonup B)$ which extends properly each partial injection $p : A \rightharpoonup B$ whose domain does not exhaust A and whose range does not exhaust B ; any fixed point of that mapping then will either be defined on all of A , witnessing that $A \leq_c B$, or its image will exhaust B , witnessing that $B \leq_c A$.

In detail, let

$$\varepsilon_A : \mathcal{P}(A) \rightarrow A, \quad \varepsilon_B : \mathcal{P}(B) \rightarrow B$$

be choice functions on A and B , supplied by the Axiom of Choice, and define π on $(A \rightharpoonup B)$ as follows:

$$\pi(p) = \begin{cases} p \cup \{(\varepsilon_A(A \setminus \text{Domain}(p)), \varepsilon_B(B \setminus \text{Image}(p)))\}, & \text{if } \text{Domain}(p) \subsetneq A \text{ \& Image}(p) \subsetneq B, \\ p, & \text{otherwise.} \end{cases}$$

This mapping is obviously expansive, and so by the Fixed Point Theorem 7.35, it has a fixed point $p^* : A \rightarrow B$, for which

$$\text{either } \text{Domain}(p^*) = A \text{ or } \text{Image}(p^*) = B.$$

CASE 1, $\text{Domain}(p^*) = A$. Now $p^* : A \rightarrow B$ is an injection of A into B , and so $A \leq_c B$.

CASE 2, $\text{Image}(p^*) = B$. In this case, p^* is a bijection of $\text{Domain}(p^*)$ with B , and so $B =_c \text{Domain}(p^*) \subseteq A$ and hence $B \leq_c A$.

(2) $\implies$ (3). By the Cardinal Comparability Hypothesis, either $A \leq_c h(A)$ or $h(A) \leq_c A$, where $h(A)$ is the Hartogs set for A ; but $h(A) \not\leq_c A$ by Hartogs' Theorem 7.34, and so $A \leq_c h(A)$; in addition, $h(A)$ is well orderable, and hence A is also wellorderable by Exercise 7.3.

(3) $\implies$ (1). If $\leq$ is a wellordering of A , then the partial function

$$\varepsilon(X) = \text{the } \leq\text{-least member of } X$$

is a choice function for A . ⊣

8.10. But is it true? (2) The import of this theorem is that if we accept the basic, constructive first six axioms of Zermelo, then the Axiom of Choice, the Hypothesis of Cardinal Comparability and the Wellordering Theorem express in three different ways the same set theoretic principle. No doubt, the Axiom of Choice is the most direct and intuitive formulation of this principle, the one which makes it most obvious that it is true. The Cardinal Comparability Hypothesis is certainly easy to understand and plausible, but few would propose it as an axiom: it has the feel of a mathematical claim which ought to be proved. Finally, the Wellordering Theorem is crystal clear in its meaning and it gives a mechanism for making choices which “explains” in some way the Axiom of Choice; but far from being obvious, it raises a flag of caution. For example, what does a wellordering of the powerset of the natural numbers $\mathcal{P}(\mathbb{N})$ look like? Without some thought, it is not even obvious that $\mathcal{P}(\mathbb{N})$ admits linear orderings (see Problem x8.9). It is quite difficult to imagine the structure of the beast, and this naturally casts doubt on the truth of the axiom which implies its existence. It is hardly surprising that the commotion about the Axiom of Choice was caused by Zermelo's proof of the implication (1) $\implies$ (3), whose conclusion is still thought by many to be counterintuitive.

In addition to the Cardinal Comparability Hypothesis and the Wellordering Theorem which are fundamental for the development of set theory, the Axiom of Choice is equivalent with a host of other propositions which are important in other areas of mathematics. We include here just two of them, which are closest to our subject and easy to prove by the methods we have been studying, but there are many others.¹⁹

¹⁹Kenneth Hoffman once declared in a class that the Tychonoff Theorem of general topology is “obviously” equivalent to the Axiom of Choice, “since all fishy, general principles are equivalent to AC”.

8.11. Theorem. *The Axiom of Choice is equivalent with the following two propositions:*

(1) **Maximal Chain Principle:** *Every poset P has a chain $S \subseteq P$ which is maximal, in the sense that for every other chain S' , $S \subseteq S' \implies S = S'$.*

(2) **Zorn's Lemma:** *If every chain in a poset P has an upper bound, then P has at least one maximal element.*

PROOF. $\mathbf{AC} \implies (1)$. Assume $\mathbf{AC}$, let $(\text{Chains}(P), \subseteq)$ be the poset of all chains in P , which is inductive by Proposition 6.14, and assume towards a contradiction that there is no maximal chain, so that by $\mathbf{AC}$,

$$(\forall S \in \text{Chains}(P))(\exists S' \in \text{Chains}(P))[S \subsetneq S'];$$

now $\mathbf{AC}$ gives us a mapping $\pi : \text{Chains}(P) \rightarrow \text{Chains}(P)$ which is expansive but has no fixed point, contradicting the Fixed Point Theorem 7.35.

(1) $\implies$ (2). If S is a maximal chain in P and M is an upper bound of S , both guaranteed by the hypotheses, then M is maximal in P —because if $M <_P M'$, then $S \cup \{M'\}$ would be a chain which extends S strictly.

(2) $\implies \mathbf{AC}$. For any two sets A, B , consider the poset $(A \rightarrow B)$ of partial injections. This is inductive, by Exercise 6.17, and so every chain has an upper bound in it; by Zorn's Lemma then, $(A \rightarrow B)$ has a maximal element f , which (as in the proof of (1) $\implies$ (2) in Theorem 8.9) establishes that either $A \leq_c B$ or $B \leq_c A$, which in turn implies $\mathbf{AC}$. $\dashv$

We now consider two easy corollaries of the Axiom of Choice which express simpler principles of choice.

8.12. Countable Principle of Choice, $\mathbf{AC}_{\mathbb{N}}$. For each set B and each binary relation $P \subseteq \mathbb{N} \times B$ between natural numbers and members of B ,

$$(\forall n \in \mathbb{N})(\exists y \in B)P(n, y) \implies (\exists f : \mathbb{N} \rightarrow B)(\forall n \in \mathbb{N})P(n, f(n)).$$

8.13. (VII) Axiom of Dependent Choices, $\mathbf{DC}$. For each set A and each relation $P \subseteq A \times A$,

$$\begin{aligned} a \in A \ \& \ (\forall x \in A)(\exists y \in A)P(x, y) \\ \implies & (\exists f : \mathbb{N} \rightarrow A)[f(0) = a \ \& \ (\forall n \in \mathbb{N})P(f(n), f(n+1))]. \end{aligned}$$

In contrast to the full Axiom of Choice which demands the existence of choice functions $f : A \rightarrow B$ for arbitrary A, B , the Countable Principle of Choice $\mathbf{AC}_{\mathbb{N}}$ justifies only a sequence of independent choices from an arbitrary set B which successively satisfy the conditions

$$P(0, f(0)), P(1, f(1)), P(2, f(2)), \dots$$

The Axiom of Dependent Choices $\mathbf{DC}$ also justifies only a sequence of choices, where, however, each of them may depend on the previous one, since they must now satisfy the conditions

$$P(f(0), f(1)), P(f(1), f(2)), P(f(2), f(3)), \dots$$

It is easily equivalent to the following, seemingly stronger principle which allows each choice to depend on all the preceding ones.

8.14. Proposition. *The Axiom of Dependent Choices is equivalent to the following proposition: for every set A and every relation $P \subseteq A^* \times A$ between strings from A and members of A ,*

$$(\forall u \in A^*)(\exists x \in A)P(u, x) \implies (\exists f : \mathbb{N} \rightarrow A)(\forall n)P(\overline{f}(n), f(n)).$$

PROOF. The implication from this version of **DC** to the “official” one is easy and we leave it for an exercise. Assuming now **DC** and the hypothesis of the seemingly stronger version, define on A^* the relation

$$Q(u, v) \iff_{\text{df}} (\exists x \in A)[v = u \star \langle x \rangle \& P(u, x)];$$

we obviously have $(\forall u \in A^*)(\exists v \in A^*)Q(u, v)$, **DC** gives us a function $g : \mathbb{N} \rightarrow A^*$ such that $g(0) = \emptyset$ and $(\forall n)Q(g(n), g(n+1))$, and the function we need is $f = \bigcup g$, for which $f(n) = g(n+1)(n)$ and $\overline{f}(n) = g(n)$. $\dashv$

8.15. Exercise. *Show the direction of Proposition 8.14 omitted in the given proof.*

8.16. Theorem. (1) *The Axiom of Choice implies the Axiom of Dependent Choices.*

(2) *The Axiom of Dependent Choices implies the Countable Principle of Choice.*

PROOF. (1) Let $\varepsilon : \mathcal{P}(A) \setminus \{\emptyset\} \rightarrow A$ be a choice function for A and assume the hypothesis of **DC**. The function $f : \mathbb{N} \rightarrow A$ that we need for the conclusion is defined by the recursion

$$\begin{aligned} f(0) &= a, \\ f(n+1) &= \varepsilon(\{y \in A \mid P(f(n), y)\}). \end{aligned}$$

(2) Assume the hypothesis of the Countable Principle of Choice, let $A = \mathbb{N} \times B$, let $a = (0, b)$ where $b \in B$ is any point satisfying $P(0, b)$, and define on A the relation

$$Q((n, x), (m, y)) \iff_{\text{df}} m = n + 1 \& P(m, y).$$

The function $f : \mathbb{N} \rightarrow \mathbb{N} \times B$ supplied by **DC** for this a and Q takes pairs as values, so $f(n) = (g(n), h(n))$, $g(0) = 0$, $h(0) = b$ for suitable functions g , h , and for every n , $g(n+1) = g(n) + 1$, $P(g(n+1), h(n+1))$. It follows that for every n , $g(n) = n$ and $P(n, h(n))$, as required by the conclusion of the Countable Principle of Choice. $\dashv$

We need a definition to formulate the most useful version of the Axiom of Dependent Choices.

8.17. Definition. A graph $(G, \rightarrow_G)$ is **grounded** or **well founded** if every non-empty subset of G has a *minimal member*, i.e.,

$$\text{if } \emptyset \neq X \subseteq G, \text{ then } (\exists m \in X)(\forall x \in X)[m \not\rightarrow_G x]. \quad (8-3)$$

A poset $(P, \leq)$ is **grounded** if the associated “inverse strict graph” $(P, >)$ is grounded, which means that for every X ,

$$\text{if } \emptyset \neq X \subseteq P, \text{ then } (\exists m \in X)(\forall x \in X)[x \leq m \implies x = m]. \quad (8-4)$$

8.18. Exercise. Prove that a linear ordering $(P, \leq)$ is grounded if and only if it is a wellordering.

8.19. Proposition. The Axiom of Dependent Choices is equivalent to the following proposition: a graph G is grounded if and only if it has no **infinite, descending chains**, i.e., there exists no function $f : \mathbb{N} \rightarrow G$ such that for all n , $f(n) \rightarrow_G f(n+1)$,

$$f(0) \rightarrow_G f(1) \rightarrow_G f(2) \rightarrow_G \dots$$

PROOF. First assume **DC**. If $f : \mathbb{N} \rightarrow G$ is an infinite, descending chain, then the set $\{f(n) \mid n \in \mathbb{N}\}$ has no minimal element, so G is not grounded. Conversely, if G has a non-empty subset X with no minimal element, then $(\forall x \in X)(\exists y \in X)[x \rightarrow_G y]$, and then **DC** gives us an infinite descending chain, starting with some arbitrary $a \in X$.

Assume now that every graph which has no infinite descending chains is grounded and the hypotheses $a \in A$ and

$$(\forall x \in A)(\exists y \in A)P(x, y)$$

of **DC** holds. Consider the graph $(A, \rightarrow_A)$ where

$$x \rightarrow_A y \iff_{\text{df}} P(x, y) \quad (x, y \in A).$$

The conclusion of **DC** is exactly the statement that $(A, \rightarrow_A)$ has an infinite descending chain which starts with a , so if it fails, there must exist some minimal $m \in A$; this means precisely that $(\forall y \in A)\neg P(m, y)$, which contradicts the hypothesis of **DC**. $\dashv$

Grounded graphs have many of the properties of well ordered sets: in particular, we can prove propositions by induction and define functions by recursion over them, cf. Problems **x8.10**, **x8.11** and Theorem **11.5**. The easy direction of this result makes **DC** particularly useful in studying them, as it is often simpler to verify that a given graph G has no infinite descending chains than to prove directly that it is grounded.

8.20. But is it true? (3) We have remarked that before it was formulated precisely by Zermelo, the Axiom of Choice had been used many times “silently” in classical mathematics, and in particular in analysis. *These classical applications, however, can all be justified on the basis of the Axiom of Dependent Choices*—in fact most of them need only the weaker Countable Principle of Choice. This will become clear in Chapter **10** and Appendix **A**. Zermelo assumed the full Axiom of Choice because it is a natural hypothesis in the context of Cantor’s set theory; because it is needed in the proofs of the Wellordering Theorem and the Cardinal Comparability Hypothesis; and because it is indispensable for the development of cardinal arithmetic. This difference between the choice principles needed for classical mathematics and those required by Cantor’s new theory of sets explains in part the strident reaction to the axioms of Zermelo by the distinguished analysts of his time (including the great Borel), who had used choice principles routinely in their work—and continued using them, as they denounced general set theory and

called it an illusion: in the context of 19th century classical analysis, the Axiom of Dependent Choices is natural and necessary, while the full Axiom of Choice is unnecessary and even has some counterintuitive consequences, including certainly the Wellordering Theorem.

We should also mention here that even in general set theory where the full Axiom of Choice is routinely accepted as obvious, many of the basic theorems do not need it, and in particular *all the results of Chapter 2 can be based axiomatically on the Axiom of Dependent Choices*. Notice also that we proved all the basic facts about well ordered sets in the preceding chapter with no appeal to choice principles whatsoever. For this reason, we will deviate technically from Zermelo and we will put in our basic system the Axiom of Dependent Choices instead of the full Axiom of Choice. “Technically”, because we adopt the view that there is no doubt about the truth of the Axiom of Choice and we will never hesitate to appeal to it when it is needed: we will simply include it (discreetly) among the hypotheses.

8.21. The axiomatic theory ZDC. The axiomatic system **ZDC** comprises the constructive axioms (I) – (VI) of Chapter 3 and the Axiom (VII) of Dependent Choices 8.13, symbolically

$$\mathbf{ZDC} = (\mathbf{I}) - (\mathbf{VI}) + \mathbf{DC} = (\mathbf{I}) - (\mathbf{VII}).$$

From now on and until Chapter 11, we will use in proofs the axioms of **ZDC** without explicit mention. When the Axiom of Choice is required, we will make a note of the fact by annotating the relevant proposition with the mark (AC). In Chapter 11 we will complete our axiomatization by adding to **ZDC** the *Axiom of Replacement*.

8.22. Consistency and independence results. Could we settle the controversy about the Axiom of Choice by simply *proving* or *refuting* AC from the constructive axioms (I) – (VI)? Neither possibility seems likely. On the one hand, AC is probably true, as are axioms (I) – (VI), and we cannot refute a true statement on the basis of true assumptions. On the other hand, AC appears to be a genuinely new set theoretic principle, and we cannot expect to prove it from the other ones, by logic alone. As a matter of fact, it can be shown rigorously that the Axiom of Choice can neither be proved nor refuted from axioms (I) – (VI).

The most direct way to show that a certain proposition ϕ cannot be proved in a certain axiomatic system **T** is to produce a **model** of **T** in which ϕ is false. Consider the classical problem about plane Euclidean geometry, whether the *Parallel Axiom*²⁰ can be deduced or not from the others. To show that it cannot, we declare that by “plane” we will mean the two-dimensional sphere, the surface of the unit ball, and by “line” we will mean any great circle on the

²⁰The Parallel (fifth) Axiom of Euclid is equivalent to the assertion that given a line L and a point P not on it, there exists exactly one line L' through P and having no points in common with L .

sphere. The remaining primitive notions of plane Euclidean geometry can be defined naturally in this interpretation, and it is not hard to verify that the basic, simple axioms of Euclid are true with these definitions; thus, we have a model of plane geometry in which the Parallel Axiom fails, simply because any two great circles intersect. We conclude that the Parallel Axiom cannot be proved from the others “by logic alone”, because then it would be true in every interpretation which makes the other axioms true, and we have found one where it is false.

To define a model for an axiomatic theory, in general, one needs to specify a domain of objects and interpret on it the primitives of the theory, so that the axioms are true. For a theory about sets, this means we must define *sethood* and *membership* on some domain, and we must also identify which conditions and operations on the domain will be considered *definite*. Now, **ZDC** is a very strong theory, and models for it do not come cheap; we will study some very special ones (“set universes”) in Appendix **B**, but the most interesting constructions require delicate methods from *mathematical logic* which are outside the scope of these Notes. Here we will just state and discuss some of the many famous consistency and independence results of the subject as they become relevant in what follows.

We have assumed at the outset, in **3.6**, that our theory has a model, the standard universe of objects $\mathcal{W}$, in which axioms **(I) – (VI)** (at least) are true. This assumption is natural and even necessary if our lives as set theorists are to have any meaning, but it is not included among the axioms of **ZDC** or any of the stronger theories we will consider.²¹ It is almost never needed, except when we assert the existence of models of extensions of **ZDC**: to construct those, we have to start with something, and that is always the assumed, standard model of our theory.

8.23. Proviso for model existence assertions. Without further mention, all claims in these Notes of existence of models, consistency of theories and independence of propositions are based on the existence of a model which satisfies axioms **(I) – (VI)** and **(VIII)**, the *Axiom of Replacement*, which we will introduce in Chapter **11**.

8.24. The consistency of the Axiom of Choice (Gödel, 1939). *Zermelo’s theory **ZDC** + AC with the full Axiom of Choice has a model*, and hence **(I) – (VI)** do not refute AC, or AC is *consistent with (I) – (VI)*. Gödel’s famous model *L* of *constructible sets* has many more canonical properties and it witnesses the consistency of AC with theories much stronger than **(I) – (VI)**. We will come back to it on several occasions.

²¹In fact it is not possible to assume such an axiom: adding the existence of a model of **ZDC** to the axioms of **ZDC** creates a new and stronger theory **ZDC’** and the further problem whether **ZDC’** has a model. In the most famous result of Mathematical Logic, Gödel proved (rigorously) that this conundrum cannot be avoided: *there exists no axiomatic theory (consistent and worth studying) which includes among its axioms or theorems the assertion that it possesses a model*.

8.25. The independence of the Axiom of Choice (Fraenkel-Mostowski, 1939, Cohen, 1963). *Each of the theories*

$$(I) - (VI) + \neg AC_{\mathbb{N}}, (I) - (VI) + AC_{\mathbb{N}} + \neg DC, ZDC + \neg AC$$

has a model. This means that we cannot prove $AC_{\mathbb{N}}$ from the constructive axioms (I) – (VI), we cannot prove DC from the constructive axioms and $AC_{\mathbb{N}}$, and we cannot prove AC in ZDC : each of these three choice principles is stronger than the preceding ones. The early model constructions of Fraenkel and Mostowski either contained atoms or had some other, technical defects which limited the possibility of generalizing them. Cohen constructed his models by his famous *forcing method*, which he (and others) also used to establish many more unprovability results. We will refer to it several times in the remainder of these Notes.

Problems for Chapter 8

Let us call two propositions ϕ and ψ **constructively equivalent** if their equivalence $\phi \iff \psi$ can be established on the basis of the constructive axioms (I) – (VI), i.e., without appealing to any choice principle whatsoever.

x8.1. Prove the Axiom of Choice (8-1) for finite A .

x8.2. The Axiom of Choice is constructively equivalent with the following proposition: for every $A \neq \emptyset$ and every $f : A \rightarrow B$, there exists some $g : B \rightarrow A$ such that for all $x \in A$, $f(g(f(x))) = f(x)$.

x8.3. The Axiom of Choice is constructively equivalent with the following proposition: for each I and each indexed family of sets $(i \mapsto A_i)$ on I ,

$$(\forall i \in I)[A_i \neq \emptyset] \implies \prod_{i \in I} A_i \neq \emptyset.$$

The Countable Principle of Choice is constructively equivalent with the proposition: for every sequence of sets $(n \mapsto A_n)$, $n \in \mathbb{N}$,

$$(\forall n \in \mathbb{N})[A_n \neq \emptyset] \implies \prod_{n \in \mathbb{N}} A_n \neq \emptyset.$$

Combined with Problem **x5.28**, the next problem gives a formulation of the Countable Principle of Choice $AC_{\mathbb{N}}$ directly in terms of the membership relation, with no reference to $\mathbb{N}$ or the concept of “function”.

x8.4. The Countable Principle of Choice $AC_{\mathbb{N}}$ is constructively equivalent with the following proposition: every countable, infinite family $\mathcal{E}$ of non-empty and pairwise disjoint sets admits a choice set.

In the next four problems we establish that the Axiom of Dependent Choices is equivalent with several seemingly weaker principles of choice.

***x8.5.** The Axiom of Dependent Choices is constructively equivalent with the following proposition: for every non-empty A and every relation $P \subseteq A \times A$,

$$\text{if } (\forall x \in A)(\exists y \in A)P(x, y), \\ \text{then } (\exists B \subseteq A)[B \neq \emptyset \ \& \ (\exists f : B \rightarrow B)(\forall x \in B)P(x, f(x))].$$

***x8.6.** The Axiom of Dependent Choices is constructively equivalent with the following proposition: for every relation $P \subseteq A \times A$ and $a \in A$,

$$\text{if } (\forall x \in A)(\exists y \in A)P(x, y), \\ \text{then } (\exists B \subseteq A)[a \in B \ \& \ (\exists f : B \rightarrow B)(\forall x \in B)P(x, f(x))].$$

***x8.7.** The Axiom of Dependent Choices is constructively equivalent with the following proposition: a poset P is grounded if and only if it has no **infinite, descending chains**, i.e., if for every $f : \mathbb{N} \rightarrow P$,

$$(\forall n \in \mathbb{N})[f(n+1) \leq f(n)] \implies (\exists n)[f(n+1) = f(n)].$$

It is also possible to formulate the Axiom of Dependent Choices directly in terms of the membership relation, but not in a very pretty manner.

***x8.8.** Prove that the following proposition is constructively equivalent with the Axiom of Dependent Choices: for every set A and every binary definite condition P ,

$$\text{if } \left[\emptyset \in A \ \& \ (\forall u, v \in A)[P(u, v) \implies (\exists! x)[v = u \cup \{x\}]] \right. \\ \left. \ \& \ (\forall u \in A)(\exists v \in A)P(u, v) \right], \\ \text{then } (\exists B \subseteq A)[\emptyset \in B \ \& \ (\forall u \in B)(\exists! v \in B)P(u, v)].$$

***x8.9.** Prove that the following, lexicographic ordering on $(\mathbb{N} \rightarrow \mathbb{N})$ is indeed a linear ordering but not a wellordering:

$$f \leq g \iff_{\text{df}} f = g \vee (\exists n \in \mathbb{N})[(\forall i < n)[f(i) = g(i)] \ \& \ f(n) < g(n)].$$

Infer that $\mathcal{P}(\mathbb{N})$ admits a linear ordering.

x8.10. Grounded induction. For every grounded graph G and each unary definite condition P ,

$$\text{if } (\forall x \in G)[(\forall y)(x \rightarrow_G y \implies P(y)) \implies P(x)], \text{ then } (\forall x \in G)P(x).$$

***x8.11.** For every grounded graph G and every function

$$h : (G \multimap E) \times G \rightarrow E,$$

there exists a unique (total) function $f : G \rightarrow E$ which satisfies the identity

$$f(x) = h(f \upharpoonright \{y \in G \mid x \rightarrow_G y\}, x) \quad (x \in G).$$

HINT. Rework the proof of Theorem 7.24, using functions

$$\sigma_t : \{x \in G \mid t \Rightarrow_G x\} \rightarrow E$$

defined on “initial segments” of the transitive closure $\Rightarrow_G$ of $\rightarrow_G$.